Configuration

- Hadoop was configured on WSL2 on a windows machine.
- Hive was installed upon that.
- Engine used was mapreduce (despite of given warnings, it was not changed to tez)

Data creation:

This python script was used to create the data

```python
import pandas as pd
import random
from datetime import datetime, timedelta
import os

# Define output directory
output_dir = '/home/hasnain_unix/myData/raw_data'
os.makedirs(output_dir, exist_ok=True)

# ------------------------- #
# Generate User Activity Logs
# ------------------------- #

# Configuration
start_date = datetime(2023, 9, 1)
num_days = 7
actions = ['play', 'pause', 'skip', 'forward']
devices = ['mobile', 'desktop', 'tablet']
regions = ['US', 'EU', 'APAC']
user_ids = range(100, 200)
content_ids = range(1000, 1011)

# Generate data for each day
for day in range(num_days):
    current_date = start_date + timedelta(days=day)
    rows = []

    for _ in range(random.randint(20, 30)):
        row = {
            'user_id': random.choice(user_ids),
            'content_id': random.choice(content_ids),
            'action': random.choice(actions),
            'timestamp': current_date.strftime("%Y-%m-%d %H:%M:%S"),
```

```
            'device': random.choice(devices),
            'region': random.choice(regions),
            'session_id': f"S{random.randint(1000, 9999)}"
        }
        rows.append(row)

    # Convert to DataFrame
    df = pd.DataFrame(rows)

    # Save to CSV
    file_path = os.path.join(output_dir, f"{current_date.strftime('%Y-%m-%d')}.csv")
    df.to_csv(file_path, index=False)
    print(f"Generated file: {file_path}")

# ------------------------- #
# Generate Content Metadata
# ------------------------- #

metadata = [
    [1000, 'Summer Vibes', 'Pop', 180, 'DJ Alpha'],
    [1001, 'Rock Anthem', 'Rock', 240, 'The Beats'],
    [1002, 'Morning Podcast', 'Podcast', 3600, 'Talk Show'],
    [1003, 'Global News', 'News', 900, 'Anchor X'],
    [1004, 'Jazz Classics', 'Jazz', 300, 'Jazz Band'],
    [1005, 'Workout Hits', 'Pop', 200, 'Fit DJ'],
    [1006, 'Evening Relaxation', 'Jazz', 600, 'Calm Notes'],
    [1007, 'Chill Beats', 'Jazz', 400, 'Smooth Artist']
]

# Convert to DataFrame
df_metadata = pd.DataFrame(metadata, columns=['content_id', 'title', 'category', 'length', 'artist'])

# Save metadata to CSV
metadata_file_path = os.path.join(output_dir, "metadata.csv")
df_metadata.to_csv(metadata_file_path, index=False)
print(f"Generated metadata file: {metadata_file_path}")

```

Following script was used for data ingestion

```bash
#!/bin/bash

# Check if the date parameter is provided
```

```
if [ -z "$1" ]; then
  echo "Usage: $0 <YYYY-MM-DD>"
  exit 1
fi

# Extract year, month, and day from the input date
DATE=$1
YEAR=$(echo $DATE | cut -d'-' -f1)
MONTH=$(echo $DATE | cut -d'-' -f2)
DAY=$(echo $DATE | cut -d'-' -f3)

# Local paths (modify according to your actual data directory)
LOCAL_DATA_DIR="/local/path/to/raw_data"
USER_LOG_FILE="${LOCAL_DATA_DIR}/${DATE}.csv"
METADATA_FILE="${LOCAL_DATA_DIR}/metadata.csv"

# Correct HDFS directories for Hive partitioning
HDFS_LOG_DIR="/raw/logs/year=${YEAR}/month=${MONTH}/day=${DAY}"
HDFS_METADATA_DIR="/raw/metadata"

# ---------------------- #
# 1. Upload User Log File
# ---------------------- #

# Create the HDFS directory for logs (partitioned by date)
echo "Creating HDFS directory: $HDFS_LOG_DIR"
hdfs dfs -mkdir -p "$HDFS_LOG_DIR"

# Upload the log file to HDFS
echo "Uploading $USER_LOG_FILE to $HDFS_LOG_DIR"
hdfs dfs -put -f "$USER_LOG_FILE" "$HDFS_LOG_DIR/"

# ------------------------- #
# 2. Upload Metadata (Once)
# ------------------------- #

# Check if metadata already exists in HDFS
if ! hdfs dfs -test -e "$HDFS_METADATA_DIR/metadata.csv"; then
   echo "Uploading metadata to $HDFS_METADATA_DIR"
   hdfs dfs -mkdir -p "$HDFS_METADATA_DIR"
   hdfs dfs -put "$METADATA_FILE" "$HDFS_METADATA_DIR/"
else
   echo "Metadata already exists in HDFS: $HDFS_METADATA_DIR"
fi
```

echo "Ingestion completed for date: $DATE"

```
## QUERIES AND RESULTS

```
No Stats for de@raw_user_logs, Columns: user_id, content_id, action, session_id, region, device, timestamp
25/03/12 23:08:05 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN optimizer.SimpleFetchOptimizer: Table de@raw_user_logs is external table, falling back to filesystem scan.
```

raw_user_logs.user_id	raw_user_logs.content_id	raw_user_logs.action	raw_user_logs.timestamp	raw_user_logs.device	raw_user_logs.region	raw_user_logs.session_id	raw_user_logs.year	raw_user_logs.month	raw_user_logs.day
NULL	NULL	action	timestamp	device	region	session_id	2023	09	01
137	1002	forward	2023-09-01 00:00:00	desktop	EU	S2283	2023	09	01
173	1000	pause	2023-09-01 00:00:00	desktop	EU	S4629	2023	09	01
175	1001	skip	2023-09-01 00:00:00	mobile	US	S4851	2023	09	01
149	1001	forward	2023-09-01 00:00:00	tablet	US	S5080	2023	09	01
105	1004	skip	2023-09-01 00:00:00	tablet	EU	S9929	2023	09	01
126	1002	skip	2023-09-01 00:00:00	tablet	US	S8017	2023	09	01
137	1009	forward	2023-09-01 00:00:00	tablet	APAC	S2611	2023	09	01
118	1009	forward	2023-09-01 00:00:00	tablet	US	S4360	2023	09	01
124	1010	play	2023-09-01 00:00:00	tablet	APAC	S6675	2023	09	01

STAR SCHEMA TABLES

Data was dumped in to this from raw_logs table in hive

```
CREATE EXTERNAL TABLE IF NOT EXISTS fact_user_actions (
    user_id INT,
    content_id INT,
    action STRING,
    timestamp TIMESTAMP,
    device STRING,
    region STRING,
    session_id STRING
)
PARTITIONED BY (year STRING, month STRING, day STRING)
STORED AS PARQUET
LOCATION '/warehouse/fact_user_actions';

CREATE EXTERNAL TABLE IF NOT EXISTS dim_content (
    content_id INT,
    title STRING,
    category STRING,
    length INT,
    artist STRING
)
STORED AS PARQUET
LOCATION '/warehouse/dim_content';
```
```

Flushing data into STAR schema

```
. . . . . . . . >     artist
. . . . . . . . > FROM raw_content_metadata;
25/03/12 23:11:35 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN calcite.RelOptHiveTable: No Stats for de@raw_content_metadata, Columns: content_id, artist, length, title, category
No Stats for de@raw_content_metadata, Columns: content_id, artist, length, title, category
25/03/12 23:11:35 [HiveServer2-Background-Pool: Thread-154]: WARN ql.Driver: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. tez) or using H
ive 1.X releases.
Query ID = hasnain_unix_20250312231135_5345d9c6-124f-42b6-92b5-ca27fc8422ad
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks determined at compile time: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
WARN  : Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. tez) or using Hive 1.X releases.
25/03/12 23:11:37 [HiveServer2-Background-Pool: Thread-154]: WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to re
medy this.
Starting Job = job_1741800879035_0001, Tracking URL = http://DESKTOP-8BL3MIG.:8088/proxy/application_1741800879035_0001/
Kill Command = /home/hasnain_unix/hadoop/bin/mapred job  -kill job_1741800879035_0001
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
25/03/12 23:11:47 [HiveServer2-Background-Pool: Thread-154]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.TaskCounter instead
2025-03-12 23:11:47,682 Stage-1 map = 0%,  reduce = 0%
2025-03-12 23:11:53,462 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 2.51 sec
2025-03-12 23:11:59,653 Stage-1 map = 100%,  reduce = 100%, Cumulative CPU 5.97 sec
MapReduce Total cumulative CPU time: 5 seconds 970 msec
Ended Job = job_1741800879035_0001
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to directory hdfs://localhost:9000/warehouse/dim_content/.hive-staging_hive_2025-03-12_23-11-35_639_5795834293664484690-1/-ext-10000
Loading data to table de.dim_content
25/03/12 23:12:00 [HiveServer2-Background-Pool: Thread-154]: WARN metadata.Hive: Cannot get a table snapshot for dim_content
25/03/12 23:12:01 [HiveServer2-Background-Pool: Thread-154]: WARN metadata.Hive: Cannot get a table snapshot for dim_content
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1  Reduce: 1   Cumulative CPU: 5.97 sec   HDFS Read: 40838 HDFS Write: 2237 HDFS EC Read: 0 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 970 msec
9 rows affected (25.559 seconds)
0: jdbc:hive2://>
```

Querying the data from fact table

```
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
5/03/12 23:15:20 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN optimizer.SimpleFetchOptimizer: Table de@fact_user_actions is external table, falling back to filesystem scan.
```

fact_user_actions.user_id	fact_user_actions.content_id	fact_user_actions.action	fact_user_actions.timestamp	fact_user_actions.device	fact_user_actions.region	fact_user_actions.session_id	fact_user_act .year	fact_user_actions.month	fact_user_actions.day
124	1010	play	2023-09-01 00:00:00.0	tablet	APAC	S6675	2023	09	01
121	1004	play	2023-09-01 00:00:00.0	desktop	US	S2744	2023	09	01
116	1007	forward	2023-09-01 00:00:00.0	mobile	APAC	S7734	2023	09	01
143	1004	pause	2023-09-01 00:00:00.0	desktop	US	S4753	2023	09	01
138	1005	forward	2023-09-01 00:00:00.0	tablet	EU	S7694	2023	09	01
178	1006	play	2023-09-01 00:00:00.0	tablet	APAC	S3861	2023	09	01
189	1007	play	2023-09-01 00:00:00.0	mobile	US	S6686	2023	09	01
152	1008	forward	2023-09-01 00:00:00.0	tablet	EU	S9174	2023	09	01
162	1003	forward	2023-09-01 00:00:00.0	mobile	APAC	S9379	2023	09	01
195	1000	pause	2023-09-01 00:00:00.0	mobile	EU	S9938	2023	09	01

```
0 rows selected (0.649 seconds)
: jdbc:hive2://>
```

SELECT * FROM fact_user_actions
WHERE year='2023' AND month='09' AND day='01'
LIMIT 10;

Result

```
10 rows selected (0.649 seconds)
0: jdbc:hive2://> SELECT * FROM fact_user_actions
. . . . . . . . > WHERE year='2023' AND month='09' AND day='01'
. . . . . . . . > LIMIT 10;
25/03/12 23:16:29 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:16:29 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
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25/03/12 23:16:29 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN optimizer.SimpleFetchOptimizer: Table de@fact_user_actions is external table, falling back to filesystem scan.
```

fact_user_actions.user_id	fact_user_actions.content_id	fact_user_actions.action	fact_user_actions.timestamp	fact_user_actions.device	fact_user_actions.region	fact_user_actions.session_id	fact_user_actions.year	fact_user_actions.month	fact_user_actions.day
124	1010	play	2023-09-01 00:00:00.0	tablet	APAC	S6675	2023	09	01
121	1004	play	2023-09-01 00:00:00.0	desktop	US	S2744	2023	09	01
116	1007	forward	2023-09-01 00:00:00.0	mobile	APAC	S7734	2023	09	01
143	1004	pause	2023-09-01 00:00:00.0	desktop	US	S4753	2023	09	01
138	1005	forward	2023-09-01 00:00:00.0	tablet	EU	S7694	2023	09	01
178	1006	play	2023-09-01 00:00:00.0	tablet	APAC	S3861	2023	09	01
189	1007	play	2023-09-01 00:00:00.0	mobile	US	S6686	2023	09	01
152	1008	forward	2023-09-01 00:00:00.0	tablet	EU	S9174	2023	09	01
162	1003	forward	2023-09-01 00:00:00.0	mobile	APAC	S9379	2023	09	01
195	1000	pause	2023-09-01 00:00:00.0	mobile	EU	S9938	2023	09	01

```
10 rows selected (0.578 seconds)
0: jdbc:hive2://>
```

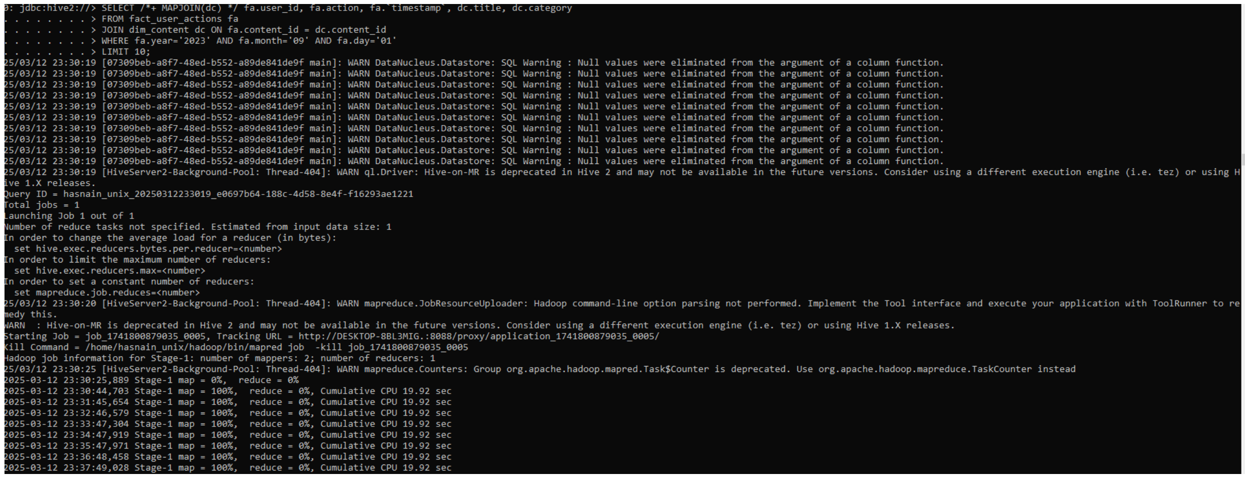
SELECT * FROM dim_content LIMIT 10;

```
10 rows selected (0.578 seconds)
0: jdbc:hive2://> SELECT * FROM dim_content LIMIT 10;
25/03/12 23:17:06 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN optimizer.SimpleFetchOptimizer: Table de@dim_content is external table, falling back to filesystem scan.
```

dim_content.content_id	dim_content.title	dim_content.category	dim_content.length	dim_content.artist
NULL	title	category	NULL	artist
1000	Summer Vibes	Pop	180	DJ Alpha
1001	Rock Anthem	Rock	240	The Beats
1002	Morning Podcast	Podcast	3600	Talk Show
1003	Global News	News	900	Anchor X
1004	Jazz Classics	Jazz	300	Jazz Band
1005	Workout Hits	Pop	200	Fit DJ
1006	Evening Relaxation	Jazz	600	Calm Notes
1007	Chill Beats	Jazz	400	Smooth Artist

```
9 rows selected (0.205 seconds)
0: jdbc:hive2://>
```

SELECT fa.user_id, fa.action, fa.timestamp, dc.title, dc.category
FROM fact_user_actions fa
JOIN dim_content dc ON fa.content_id = dc.content_id
WHERE fa.year='2023' AND fa.month='09' AND fa.day='01'
LIMIT 10;



```
0: jdbc:hive2://> SELECT /*+ MAPJOIN(dc) */ fa.user_id, fa.action, fa.`timestamp`, dc.title, dc.category
. . . . . . . . > FROM fact_user_actions fa
. . . . . . . . > JOIN dim_content dc ON fa.content_id = dc.content_id
. . . . . . . . > WHERE fa.year='2023' AND fa.month='09' AND fa.day='01'
. . . . . . . . > LIMIT 10;
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [07309beb-a8f7-48ed-b552-a89de841de9f main]: WARN DataNucleus.Datastore: SQL Warning : Null values were eliminated from the argument of a column function.
25/03/12 23:30:19 [HiveServer2-Background-Pool: Thread-404]: WARN ql.Driver: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. tez) or using Hive 1.X releases.
Query ID = hasnain_unix_20250312233019_e0697b64-188c-4d58-8e4f-f16293ae1221
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
25/03/12 23:30:20 [HiveServer2-Background-Pool: Thread-404]: WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
WARN  : Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. tez) or using Hive 1.X releases.
Starting Job = job_1741800879035_0005, Tracking URL = http://DESKTOP-8BL3MIG.:8088/proxy/application_1741800879035_0005/
Kill Command = /home/hasnain_unix/hadoop/bin/mapred job  -kill job_1741800879035_0005
Hadoop job information for Stage-1: number of mappers: 2; number of reducers: 1
25/03/12 23:30:25 [HiveServer2-Background-Pool: Thread-404]: WARN mapreduce.Counters: Group org.apache.hadoop.mapred.Task$Counter is deprecated. Use org.apache.hadoop.mapreduce.TaskCounter instead
2025-03-12 23:30:25,889 Stage-1 map = 0%,  reduce = 0%
2025-03-12 23:30:44,703 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:31:45,654 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:32:46,579 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:33:47,304 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:34:47,919 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:35:47,971 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:36:48,458 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
2025-03-12 23:37:49,028 Stage-1 map = 100%,  reduce = 0%, Cumulative CPU 19.92 sec
```

Its taking long because i have not shifted to `tez` but as visible in the logs, mapreduce is working fine.

Monthly active users by region

```
SELECT
    fa.region,
    fa.year,
    fa.month,
    COUNT(DISTINCT fa.user_id) AS active_users
FROM fact_user_actions fa
WHERE fa.year = '2023' AND fa.month = '09'
GROUP BY fa.region, fa.year, fa.month
ORDER BY active_users DESC;
```